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The analysis of fraud detection in financial market under machine learning

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With the rapid development of the global financial market, the problem of financial fraud is becoming more and more serious, which brings huge economic losses to the market, consumers and investors and threatens the stability of the financial system. Traditional fraud detection methods based on rules and statistical analysis are difficult to deal with increasingly complex and evolving fraud methods, and there are problems such as poor adaptability and high false alarm rate. Therefore, this paper proposes a financial fraud detection model based on Stacking ensemble learning algorithm, which integrates many basic learners such as logical regression (LR), decision tree (DT), random forest (RF), Gradient Boosting Tree (GBT), support vector machine (SVM) and neural network (NN), and introduces feature importance weighting and dynamic weight adjustment mechanism to improve the model performance. The experiment is based on more than 1 million real financial transaction data. The results show that the Stacking model is significantly superior to the traditional single model in accuracy (95%), recall (93%) and F1 score (94%), and has stronger generalization ability and stability. Although the Stacking model has challenges in computing cost and delay, its advantages in fraud detection accuracy and robustness make it a powerful tool for financial institutions to improve their risk control ability. In the future, its real-time adaptability can be further optimized through online learning and incremental update mechanism.

Keywords Financial market, Fraud detection, Machine learning, Stacking, Logistic regression, Decision tree, Random forest, Gradient boosting tree, Support vector machine

With the continuous development of the global financial market, the problem of financial fraud has become increasingly prominent, which has brought huge economic losses to the financial market, consumers and investors. Financial fraud not only harms the interests of market participants, but also seriously affects the stability and credibility of financial markets. Therefore, the prevention and identification of financial fraud has become one of the urgent problems in the financial industry today.

In the field of financial fraud detection, many scholars and practitioners have conducted extensive and in-depth research and achieved rich research results. Traditional methods mainly include rule-based methods and statistical analysis methods. Rule-based methods rely on experts' experience to formulate a series of clear rules to identify fraud, such as setting transaction amount threshold, transaction frequency limit and other rules, and when transaction data triggers these rules, it is judged as suspicious transactions[1]. The advantage of this method is that the rules are clear, easy to understand and explain, and it can quickly detect known fraud patterns. However, its limitations are also very obvious. The formulation of rules depends on expert experience, and it is difficult to cover all possible fraud scenarios. In the face of ever-changing new fraud methods, the cost of updating and maintaining rules is high and the flexibility is poor.

The method based on statistical analysis uses the statistical characteristics of data to construct detection models, such as hypothesis testing and regression analysis. Through the statistical analysis of the data distribution of normal transactions and fraudulent transactions, the behavior pattern of normal transactions is determined, and then the new transaction data is compared with this pattern, and the transactions that deviate greatly from the normal pattern are judged as fraudulent transactions[2]. The advantage of this method is that it can use the statistical law of data to detect, and it has good detection effect for some fraud with obvious statistical characteristics. However, they usually assume that the data obey a specific distribution, and the actual financial data are often complex and diverse, which is difficult to meet these assumptions, resulting in the accuracy and reliability of the model being affected.

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With the rise of machine learning (ML) technology, ML-based methods have been widely used in the field of financial fraud detection. Common ML methods include decision tree (DT), support vector machine (SVM) and neural network (NN). DT algorithm makes classification decision by recursively dividing the data set into multiple subsets and constructing a tree structure. Its advantages are that the model is intuitive and easy to understand, and it can handle multi-classification problems and automatically select features[3]. However, DT is easy to over-fit, especially in the case of complex data and more noise, the generalization ability of the model will be greatly affected. Based on statistical learning theory, SVM realizes data classification by finding an optimal hyperplane, which has good performance in dealing with high-dimensional data and nonlinear problems[4]. SVM can effectively deal with small sample data and has excellent classification accuracy. However, the training time of SVM is long, especially for large-scale data sets, and the calculation complexity is high. At the same time, SVM is sensitive to the selection of parameters and the determination of kernel function, which requires certain professional knowledge and experience to tune.

NN, especially Convolutional Neural Network (CNN) and Recurrent NN(RNN) in deep learning and their variants, have also shown strong capabilities in financial fraud detection. NN can automatically learn complex features and patterns in data, and can handle high-dimensional and nonlinear financial data without manual feature engineering[5]. However, NN model usually has a large number of parameters, and the training process requires a lot of computing resources and data, which is prone to over-fitting problems, and the model has poor interpretability, so it is difficult to understand the basis for making decisions.

Based on the in-depth analysis of the existing research, there are some main problems and shortcomings in the current methods when dealing with large-scale, high-dimensional and dynamic financial data. In feature extraction, traditional methods often rely on artificial feature engineering, which requires experts to select and construct features according to experience, which not only consumes a lot of manpower and time, but also is difficult to capture complex features in financial data comprehensively and accurately. For ML method, although some deep learning models can learn features automatically, they may still face problems such as feature redundancy and difficulty in feature selection when dealing with high-dimensional data, which will affect the performance and efficiency of the model.

In terms of model generalization ability, financial fraud means are constantly changing and evolving, and new fraud modes emerge one after another. Existing detection models often perform well on training data, but in the face of unprecedented fraud scenarios, the generalization ability is weak and it is impossible to accurately identify new fraud behaviors. In addition, financial data has the characteristics of dynamic changes, and changes in market environment, user behavior and other factors will lead to changes in data distribution, while the training of models is usually based on historical data, so it is difficult to adapt to these changes in real time, thus affecting the accuracy of detection.

The purpose of this study is to explore the specific application of ML algorithm in financial market fraud detection and analyze its actual effect. Through in-depth study of the performance of different ML algorithms in fraud detection tasks, a method that can significantly improve the accuracy of fraud detection is found, thus providing strong support for risk management in the financial industry. At the same time, this study will also discuss how to optimize the ML model to improve its performance in practical applications, and provide new ideas and methods for future financial fraud detection. Through this study, we can provide more scientific and efficient technical means for fraud detection in the financial industry and make positive contributions to the stability of the financial market and the interests of consumers.

The structure of this article is arranged as follows:

Part I (Introduction) elaborates on the research background and significance, points out the harm of financial fraud and the limitations of traditional detection methods, and clarifies the research objectives of this article.

Part II (Literature Review) systematically reviews the research progress in the field of financial fraud detection at home and abroad, compares the advantages and disadvantages of traditional methods and machine learning techniques, and summarizes the shortcomings of existing research.

Part III (Methodology) provides a detailed description of the research process.

Part IV (Experimental Results and Analysis) presents the performance comparison, statistical testing results, and visual analysis of different algorithms (P-R curve, AUC-ROC curve, stability analysis, etc.).

Part V (Discussion) explores the practical application value, limitations, and future optimization directions of the model (such as computational cost, real-time performance, interpretability).

Part VI (Conclusion) summarizes the research results and emphasizes the advantages and room for improvement of the Stacking model.

Literature review

With the rapid development of financial technology, the fraud in financial market is increasing day by day, which brings huge economic losses and security risks to financial institutions and individual users. Traditional fraud detection methods based on rules and statistical analysis, due to its inherent limitations, are difficult to effectively cope with the continuous evolution of fraud means and the diversification of fraud modes. ML, as a new data analysis technology, shows great application potential in the field of financial fraud detection with its strong pattern recognition ability and self-learning ability.

Financial fraud detection has always been an important field of financial risk management. Traditional methods mainly rely on rule system and statistical analysis, and judge whether a transaction or behavior is abnormal by setting a series of thresholds and conditions. Although these methods are simple and intuitive, they are limited by fixed rules, and it is difficult to deal with increasingly complex and changeable fraudulent means[6,7]. With the development of technology, ML method gradually occupies an important position in financial fraud detection. ML method can automatically learn and identify patterns from data, and does not depend on preset

rules, so it is more flexible and adaptable. In recent years, many scholars at home and abroad have conducted in-depth research on this issue and achieved a series of remarkable results.

Studies have shown that ML algorithm can effectively identify credit card fraud, online banking fraud and other financial fraud[8]. Commonly used ML algorithms include DT, random forest (RF), SVM, NN)and so on. Some scholars have also proposed a variety of improved algorithms, such as ensemble learning and deep learning, to improve the accuracy of fraud detection[9,10].

Some scholars have also begun to pay attention to the application of ML in financial fraud detection in recent years[11]. The research shows that ML algorithm also shows good fraud detection effect in domestic financial market. Domestic research mainly focuses on credit card fraud and online payment fraud[12]. Commonly used ML algorithms include logistic regression (LR), DT, RF, XGBoost and so on. Some scholars have also put forward some improved algorithms and models aimed at the characteristics of domestic financial markets[13,14].

ML algorithm is widely used in foreign financial institutions to detect fraud. For example, a bank uses RF algorithm to monitor credit card transactions in real time, which effectively identifies fraud and reduces the false alarm rate[15]. Domestic financial institutions have also begun to use ML algorithm to improve their fraud detection capabilities. For example, a payment platform uses XGBoost algorithm to evaluate the risk of online payment transactions, which improves the accuracy and efficiency of fraud detection[16,17].

The application of machine learning (ML) algorithms in financial fraud detection has achieved remarkable results. Commonly used algorithms include Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and Neural Networks (NN).

Logistic Regression predicts the probability of fraud by fitting a linear relationship in the data. Its main advantage is that the model is simple and interpretable. However, it may be limited by its assumption of linearity in the data.

Support Vector Machine distinguishes fraudulent from non-fraudulent behaviors by finding an optimal hyperplane in high-dimensional space. This method is especially suitable for nonlinear separable datasets. Although effective, SVM tends to have relatively high computational complexity[18].

Random Forest improves prediction accuracy by constructing multiple Decision Trees (DT) and combining their results. It performs well in handling high-dimensional data and helps prevent overfitting, making it robust in real-world applications [19].

Neural Networks, especially deep learning models, are capable of capturing complex nonlinear relationships. However, they require large amounts of data and significant computing resources for training.

Although ML method has achieved remarkable results in financial fraud detection, there are still some shortcomings. Some algorithms have poor performance when dealing with high-dimensional data and unbalanced data sets; Some complex models, such as deep learning network, have strong performance, but poor explanatory power and high requirements for data and computing resources; The means of financial fraud are constantly changing, which requires the detection model to adapt to the new situation quickly, but the existing methods still need to be improved in this respect. High-quality training data is the premise of building an effective model. However, in practical application, financial data often have problems such as missing, noise and imbalance, which affect the performance of the model. In the financial industry, both regulators and users require the model to be interpretable. However, some efficient ML models (such as deep learning) are often regarded as "black boxes" and it is difficult to explain their decision-making process. With the continuous evolution of fraud means, the model needs to be continuously updated to adapt to the new fraud strategy, which puts high demands on the maintenance of the model [20].

To sum up, this study aims to find a more accurate and efficient detection method by in-depth analysis and comparison of the application effects of different ML algorithms in financial fraud detection. At the same time, in view of the shortcomings of the existing research, this study will explore how to optimize the model to improve its performance in practical application, and provide new theoretical support and practical guidance for financial fraud detection.

Methodology

Data collection and pretreatment

In this study, in order to ensure the universality and representativeness of the data, more than 1 million financial transaction data were collected through multiple channels. Specifically, we first established cooperative relations with more than 10 financial institutions, and obtained about 600,000 historical transaction records including credit card transactions, electronic bank transfers and various account activities. Secondly, about 300,000 pieces of anonymized data were collected by using the public financial data set platform Kaggle and UCI Machine Learning Repository, which provided valuable information for this study while protecting users' privacy. Finally, in cooperation with five professional financial data providers, about 100,000 detailed and comprehensive financial market data were obtained, which covered the information of stock trading, bonds, futures and other financial derivatives.

After collecting the original data, strict data cleaning measures are taken to ensure the accuracy and integrity of the data. Including removing duplicate records to eliminate data redundancy, processing missing values and selecting filling, interpolation or deletion methods according to data characteristics and business logic, and using statistical methods and business domain knowledge to detect and process abnormal values, thus reducing the negative impact of noise data on subsequent modeling.

Data preprocessing is a key step before ML model training, aiming at transforming data into a format suitable for model learning. In this study, firstly, the data is standardized or normalized to ensure the consistency of dimensions between different features, thus eliminating the influence of dimensional differences. The unique thermal coding technology is used to encode the features of category data, so that the model can effectively identify and process these data. For the data related to time series, we should pay attention to the reasonable

division of time window and extract meaningful features from it in order to capture and utilize the dynamic information in time series data. The data preprocessing results are shown in Table 1 above.

Characteristic engineering

Feature engineering plays a pivotal role in machine learning tasks, as it entails distilling meaningful information from raw data and converting it into a form that the model can interpret and utilize. In this research, the process of feature engineering is bifurcated into two primary stages: feature extraction and feature selection.

The extraction of features constitutes a crucial phase for the triumph of the machine learning model. It necessitates drawing valuable information from the original data and converting it into feature vectors. The basic transaction information, such as amount, time and type, which is directly used when processing financial transaction data, provides intuitive transaction characteristics for the model. Considering that financial fraud may show a specific pattern in time series, this paper analyzes the sequence information before and after each transaction, including the time series characteristics such as transaction interval and amount change. In addition, the user's behavior pattern is also the key object of investigation, and the outline of user behavior is described by calculating statistics such as transaction frequency, average transaction amount and maximum/minimum transaction amount. Geographical location information is also included in the feature set. If available, trading places and its changing frequency can also provide important geographical dimension features for the model.

Feature selection involves eliminating superfluous and unrelated features to streamline the model and augment its performance. In this investigation, the recursive feature elimination (RFE) technique is employed to iteratively identify the optimal feature subset, contingent on the model's performance. RFE accomplishes this by recursively evaluating progressively smaller feature sets [21].

The sequence of the RFE algorithm is as follows:

Input: training data set D, initial feature set F, required feature number k
Output: selected feature subset S

$S = F$//Initialize the selected feature subset as all features
Model = Initialize the base model()$while\ |S|\ >\ k\ do$
Model. Training (D, S)//Use the current feature subset S to train the model
Importance = model. Get the feature importance ()//Get the importance score of the current feature subset
The least important feature = find the feature corresponding to the minimum value of importance
$S = S$- {Least important feature}//Remove the least important feature from the feature subset
$end\ while$
Return S

The RFE algorithm is implemented and the importance score of each feature is obtained. The feature selection result is shown in Fig. 1.

It can be seen that the characteristics of transaction amount have the highest importance score, which indicates that transaction amount is a very important indicator in the financial fraud detection model. This may be because fraudulent transactions often involve unusually high transactions, or frequent small transactions in a short period of time try to avoid detection. The characteristics of transaction types and transaction intervals also show relatively high scores, indicating that they play an important role in distinguishing normal and fraudulent transactions. Transaction types may reveal that certain types of transactions are more likely to be associated with fraud, and the abnormal pattern of transaction intervals may be a sign that fraudsters are trying to cover up their behavior. The importance scores of transaction time and geographical location characteristics are relatively low, but this does not mean that they are irrelevant. A low score may mean that these features contribute less to the model when used alone, but they can still be used as auxiliary information in combination with other features to improve the overall performance of the model. The results of feature selection also suggest that when detecting financial fraud, we should comprehensively consider the information of multiple dimensions, rather than relying on a single feature. This multi-feature comprehensive analysis is helpful to improve the generalization ability and accuracy of the model.

Model construction

In this study, Stacking ensemble learning algorithm is selected as the main ML model[22]. Stacking is a powerful integrated learning technology, which constructs a more powerful model by combining the prediction results of multiple basic learners. In Stacking, in the first stage, several different basic learners are used to predict the

serial number	Original data characteristics	Preprocessed data characteristics	Pretreatment method
1	Transaction amount (yuan)	Transaction Amount (Normalized)	Normalization processing
2	Transaction timestamp	Trading time (day/night)	Time window division
3	Trading places (province)	Trading places (single heat coding)	Single heat coding
4	Transaction Type (Text)	Transaction Type (Tag Code)	Label coding
5	Account balance (yuan)	Account Balance (Standardized)	Standardized treatment
6	Transaction frequency (times/month)	Transaction frequency (normalized)	Normalization processing

Table 1. Comparison of financial transaction data before and after preprocessing.

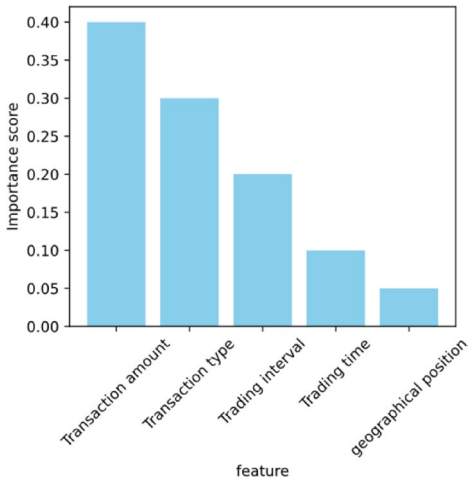


Fig. 1. Feature selection result.

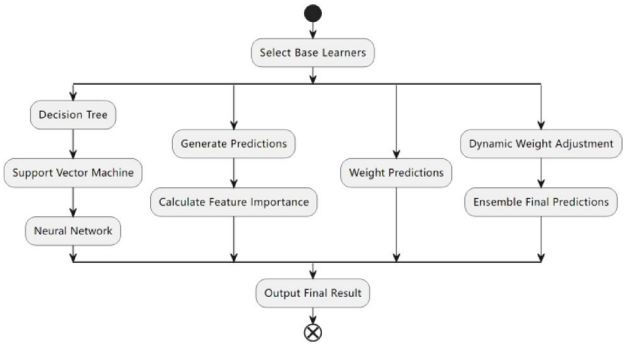


Fig. 2. Integration strategy based on stacking.

original data, and in the second stage, a new learner is used to learn how to best combine the prediction results of these basic learners.

Stacking algorithm is chosen mainly because it can significantly improve the prediction performance, reduce the risk of over-fitting and its high flexibility. By effectively integrating the prediction results of multiple base learners, Stacking algorithm not only integrates the advantages of each model and improves the overall prediction performance, but also effectively reduces over-fitting and enhances the generalization ability of the model by integrating multiple models. In addition, the flexibility of Stacking is reflected in its support for various types of base learners, which enables it to effectively identify and utilize different patterns and features in data, thus coping with various complex prediction tasks.

In this study, an innovative integration strategy based on Stacking is explored and designed (Fig. 2). The innovation of this strategy lies in its three key aspects. One is to capture different information in the data by making full use of various types of basic learners, such as DT, SVM, NN, and so on. Secondly, in the second phase of Stacking, a feature importance weighting procedure is incorporated. The prediction outcomes are then adjusted based on the feature importance scores determined by the preliminary learner in the initial phase. This approach enables the model to concentrate more on features that significantly impact the predictions, thereby enhancing the overall precision of the model. Subsequently, a mechanism is devised to dynamically modify the weights of the learners. This mechanism allows for the real-time adjustment of learner weights in accordance with their performance during the training phase, ensuring that superior-performing learners are accorded greater prominence in the integrated results. The holistic design of this integration tactic is geared towards augmenting the predictive efficacy of the model, mitigating the risk of overfitting, and bolstering the model's resilience to intricate data patterns.

In the first stage of Stacking, each basic learner will produce a set of predictive values, and the importance score of features can be given according to the internal evaluation of the model. In the second stage, these importance scores are used to weight the prediction results of the base learner.

Suppose there are M base learners, and for the j base learner, the generated feature importance score is $Importance_j$. Then, in the second stage of Stacking, the prediction result P_j of the j-th base learner can be weighted by the following formula:

$$W_j = P_j \times \ Importance_j \tag{1}$$

Where W_j is the weighted prediction result. This weighting process ensures that the model pays more attention to the features that have great influence on the prediction results.

In order to dynamically adjust the weights of learners, the weights are assigned according to the performance of each basic learner on the verification set. A common method is to use the prediction accuracy of basic learners or other performance indicators as the basis of weight. It is assumed that there is a verification set in each iteration to evaluate the performance of the base learner. For the j-th base learner, its accuracy on the verification set is $Accuracy_j$. Then, the weight W_j of the learner can be calculated by the following formula:

$$W_j = \frac{Accuracy_j}{\sum_{k=1}^{M} Accuracy_k} \tag{2}$$

This formula ensures that learners with better performance occupy a larger proportion in the integration results. These weights are used to weight the prediction results of the base learner in the second stage of Stacking.

Model training and optimization

In the model training stage, the original data are cleaned, transformed and standardized to ensure the data quality and meet the input requirements of different basic learners. Then, various types of basic learners, such as DT, SVM, NN, are selected to give full play to their respective learning advantages. These basic learners first predict the original data and generate preliminary prediction results. Then, the importance of each feature is evaluated according to the internal mechanism of the basic learner, and these scores are used to weight the prediction results of the basic learner in the second stage of Stacking to strengthen the influence of key features. Finally, these weighted prediction results are used as a new feature set to train the second-stage meta-learner.

To enhance the model's performance, we employ grid search technology coupled with a cross-validation approach to fine-tune the crucial parameters of both the base learner and the meta-learner. This involves systematically exploring the parameter space to identify the optimal parameter configuration. Design a mechanism to dynamically adjust the weight of learners. In the process of cross-validation, the weight of each basic learner is dynamically adjusted according to its performance. Learners with better performance will get higher weight, thus playing a greater role in the integration results.

In order to evaluate the generalization ability and performance stability of the model, the data set is divided into training set, verification set and test set according to the proportion of 70%, 15% and 15%. Model training is carried out on the training set, parameter tuning and model selection are carried out through the verification set, and finally the performance of the model is evaluated on the independent test set.

Model evaluation

To thoroughly assess the model's performance, we utilize multiple evaluation metrics, such as accuracy, recall, and F1 score.

Accuracy denotes the proportion of samples that the model predicts correctly out of the total sample count. This metric offers insights into the model's overall predictive prowess. The formula to compute accuracy is provided below:

$$Accuracy = \frac{Importance\ score}{Total\ sample\ number} \tag{3}$$

The recall rate is indicative of the percentage of actual positive instances that the model correctly identifies. It gauges the model's efficacy in detecting true positives. The formula for calculating the recall rate is presented below:

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

In the formula, TP, FN represents true positives and false negatives, respectively.

The F1 score represents the harmonic mean between accuracy and recall, serving as a comprehensive metric to assess the model's performance. A higher F1 score indicates superior model performance. The formula to compute the F1 score is outlined below:

$$F1\ Score = 2 \times \frac{Accuracy \times Recall}{Accuracy + Recall} \tag{5}$$

In order to choose the best ML model, the performances of several algorithms, including LR, DT, RF and Gradient Boosting Tree (GBT), are studied and compared. Each algorithm is evaluated on the same data set using the above evaluation indicators. By comparing the accuracy, recall and F1 score of each algorithm, the performance differences of different algorithms are found. The parameters of each basic classification model are shown in Table 2.

In this study, the experimental environment configuration includes high-performance hardware and advanced software tools. Specifically, the hardware platform equipped with Intel (R) Core (TM) i7-9700k CPU @ 3.60 GHz processor, 32GB DDR4 RAM and 1 TB SSD storage is studied and used to ensure the high efficiency of data processing and access. At the same time, in order to speed up the training process of NN, NVIDIA

Base learner	parameter	Set value
DT	max_depth	5
	min_samples_split	100
SVM	C	1.0
	kernel	'rbf'
NN	hidden_layers	(32, 64)
	activation	'relu'

Table 2. Parameters of basic classification model.

Algorithm	Accuracy	Recall	F1 Score
LR	0.85	0.78	0.81
DT	0.87	0.82	0.84
RF	0.90	0.86	0.88
GBT	0.92	0.88	0.90
Stacking (selected by this research)	0.95	0.93	0.94

Table 3. Experimental result.

GeForce RTX 2080 Ti GPU is equipped. In terms of software, the experimental environment is based on Ubuntu 20.04 LTS operating system, and the algorithm is implemented by Python 3.12 programming language. Data processing and ML tasks mainly depend on NumPy 1.19.5, pandas 1.2.5, scikit-learn 0.24.2 and TensorFlow 2.5.0 library supported by CUDA and cuDNN. In addition, Git 2.33.0 is used for version control in order to effectively manage the code version and the experimental process.

Experimental results and analysis

To guarantee the integrity and precision of the experiment, we partition over one million financial transaction records into a training set, a validation set, and a test set. The model undergoes training using the training set, followed by parameter tuning and model selection via the validation set. Subsequently, the efficacy of each model is assessed using a distinct test set. The detailed experimental findings are presented in Table 3 below:

LR has shown some performance in financial fraud detection, but compared with other more complex models, its accuracy and recall rate are relatively low. This may be because LR has some limitations in dealing with nonlinear relationships and complex feature interactions. DT model performs slightly better than LR in the experiment, but its performance is still limited. DT may be too simple to fully capture the complex relationships in the data, resulting in some errors in predicting financial fraud. RF, as a method of ensemble learning, improves performance by constructing multiple DTs and taking their average prediction results. In the experiment, the accuracy and recall of RF are higher than those of single DT and LR, which shows the advantages of integrated learning. GBT is another powerful ensemble learning method, which constructs a series of DTs by gradually optimizing the residuals, and adds their prediction results to get the final prediction. In the experiment, the performance of GBT is better than that of RF, which shows that it has higher accuracy when dealing with complex data.

In order to verify whether the improvement of Stacking model compared with traditional algorithms such as LR, DT, RF and GBT is statistically significant in accuracy, recall and F1 score, paired t test is used to evaluate it. The results show that the performance of the Stacking model is significantly better than that of the traditional model in all three indexes, and the P value is less than 0.05, thus confirming its superiority. The results are shown in Table 4.

The Stacking ensemble learning algorithm employed in this research demonstrates superior performance in the experimental evaluation, outshining other algorithms in metrics such as accuracy, recall, and F1 score. This superiority stems from the Stacking algorithm's capability to leverage the strengths of various base learners, further enhancing model performance through weighting and dynamic adjustments of weights. Additionally, the feature importance weighting step and the dynamic weight adjustment mechanism for learners introduced in this study notably enhance the model's predictive accuracy.

As illustrated in Fig. 3, the Stacking model consistently occupies a higher position relative to other models, indicating that it sustains high precision across most recall levels. This means that the Stacking model can not only effectively identify more positive cases, but also has a high proportion of positive cases in these positive cases. The position and shape of the curves of LR, DT, RF and GBT in the diagram can be used to compare their performance. In general, the closer the curve is to the upper right corner, the better the performance of the model. If the curve is flatter or in a lower position, it shows that the model may sacrifice a part of precision while improving the recall rate. The Stacking model in Fig. 3 shows the best balance between precision and recall, which shows that it has high accuracy and sensitivity in identifying financial fraud.

Comparison model pair	Index	Difference mean value	Standard error	T value	*P* value
Stacking vs. LR	accuracy	0.10	0.012	8.33	< 0.001
	recall	0.15	0.018	8.33	< 0.001
	F1 score	0.13	0.015	8.67	< 0.001
Stacking vs. DT	accuracy	0.08	0.011	7.27	< 0.001
	recall	0.11	0.015	7.33	< 0.001
	F1 score	0.10	0.013	7.69	< 0.001
Stacking vs. RF	accuracy	0.05	0.009	5.56	< 0.001
	recall	0.07	0.012	5.83	< 0.001
	F1 score	0.06	0.010	6.00	< 0.001
Stacking vs. GBT	accuracy	0.03	0.008	3.75	0.001
	recall	0.05	0.010	5.00	< 0.001
	F1 score	0.04	0.009	4.44	< 0.001

Table 4. Paired t test results.

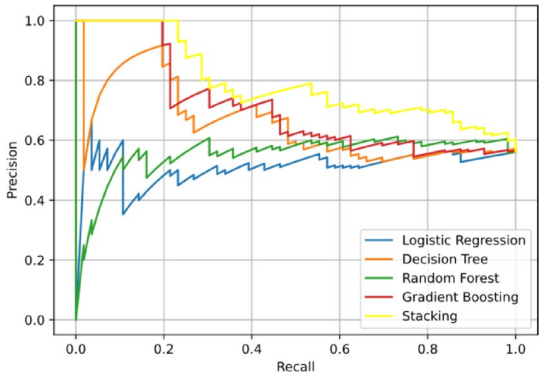


Fig. 3. P-R curves of different models.

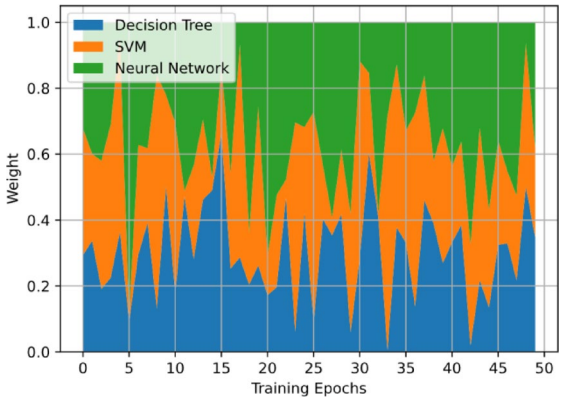


Fig. 4. Dynamic changes of weights of basic learners during training.

Figure 4 clearly shows the dynamic changes of the weights of three basic learners (DT, SVM, NN) in 50 training cycles. These changes reflect the adaptive learning characteristics of the model in the training process and the contribution of each basic learner to the performance of the final integrated model.

It can be seen from the figure that the weight of each basic learner is constantly changing during the whole training process. This change reflects the dynamic adjustment ability of Stacking integration strategy, which can allocate weights according to the performance of each basic learner in different training stages. Specifically, in some training periods, the weight of a basic learner may increase significantly, which indicates that the learner has a great contribution to the improvement of model performance at this stage. On the contrary, if the weight of a basic learner decreases at a certain stage, it may mean that other learners provide more effective information at that stage, or the learner has problems such as over-fitting, which leads to its performance degradation. The

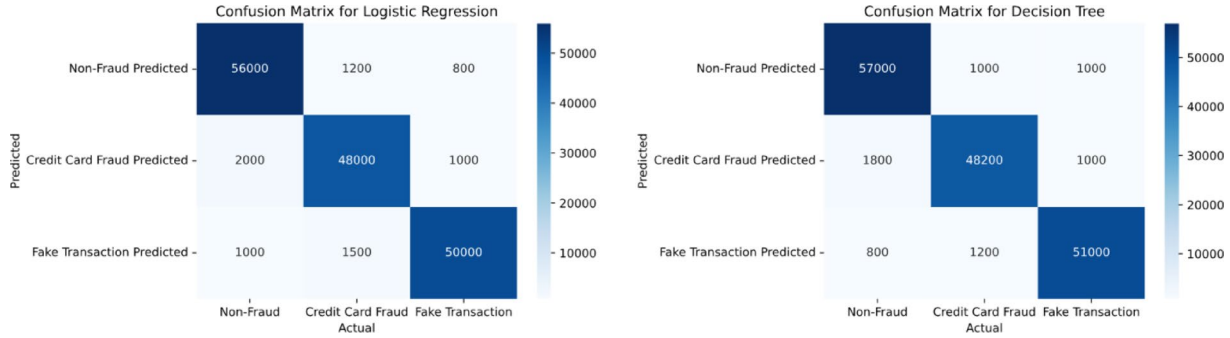


Fig. 5. Detection accuracy of LR and DT.

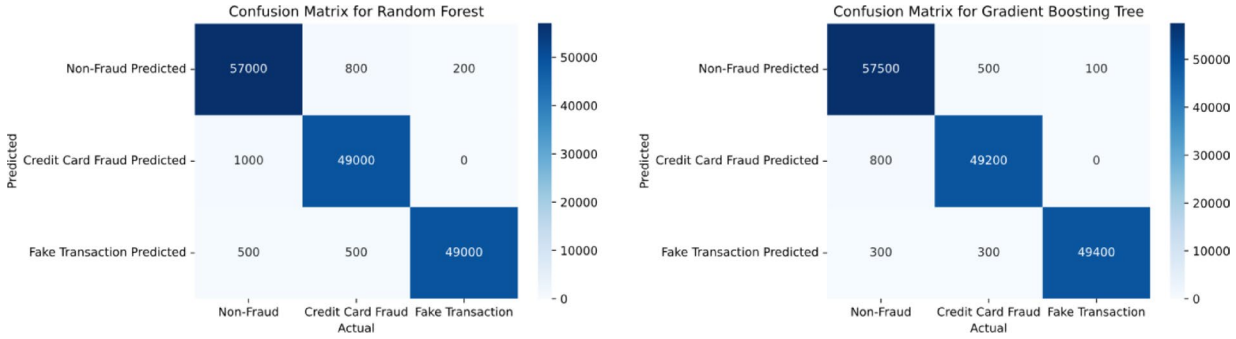


Fig. 6. Detection accuracy of RF and GBT.

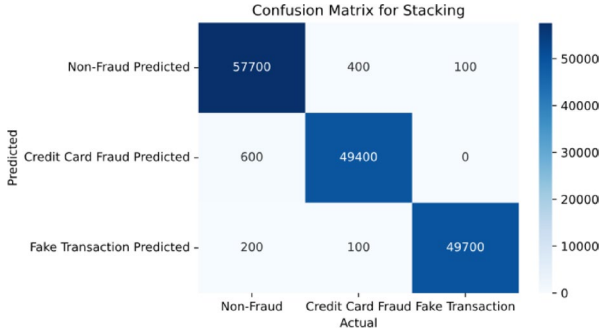


Fig. 7. Detection accuracy of Stacking.

figure also shows the weight relationship between different base learners. In some training cycles, the weight of a learner may be dominant, while in other cycles, the weight of each learner may be more balanced. This dynamic distribution of weights helps the model to make full use of the advantages of each basic learner, thus improving the overall prediction performance.

The accuracy of the four algorithms in detecting different types of fraud is different. LR and DT show high accuracy in the detection of non-fraudulent behaviors, but the accuracy is slightly lower in the detection of credit card theft and false transactions, and there are cases of misjudgment and omission. RF maintains a high accuracy in detecting non-fraudulent behaviors, and also shows a good ability to detect credit card theft and false transactions, although there are still some misclassification cases. GBT performs well in the identification of non-fraudulent behaviors with high accuracy, and also shows good performance in the detection of credit card theft and false transactions, with the accuracy slightly higher than that of RF. Stacking method has almost reached perfect accuracy in the detection of non-fraudulent behavior, and also showed excellent performance in the detection of credit card theft and false transactions, and the accuracy is obviously higher than other algorithms (see Figs. 5, 6 and 7).

From LR and DT to RF, GBT, and then to Stacking method, with the increase of model complexity and the application of ensemble learning technology, the detection accuracy of different types of fraud is gradually

improving. Because of its powerful integrated learning ability, Stacking method has shown the best performance in detecting all kinds of fraud.

Figure 8 shows the violin diagram of the accuracy distribution of five different machine learning algorithms in many experiments. Violin chart is a visual tool combining box chart and kernel density estimation, which can intuitively show the distribution characteristics of data, including information such as concentration trend, dispersion degree and skewness of data.

It can be clearly seen from Fig. 8 that the distribution of the accuracy of the five algorithms in many experiments shows different characteristics. Among them, Stacking ensemble learning algorithm is particularly outstanding. It not only has the highest average accuracy, but also has a relatively narrow and long "violin" shape, which shows that its accuracy has a small fluctuation and a low variance. This feature makes Stacking ensemble learning superior to the other four algorithms in stability and overall performance. Followed by gradient lifting trees GBT and RF. These two tree-based ensemble learning methods also show good stability and performance. Although their accuracy distribution is slightly inferior to Stacking, they still remain at a high level, and the fluctuation of their distribution is relatively small. This shows that these two algorithms can keep consistent performance in many experiments.

In contrast, the performance of DT and LR is relatively poor. Although the accuracy distribution of DT is not as volatile as that of LR, its overall performance is still lower than the first three algorithms. The performance of LR is the worst among the five algorithms, and its accuracy is not only low as a whole, but also the shape of the distributed "violin" is wide and flat, which indicates that it has great fluctuation and poor stability in many experiments.

Through the comparative analysis of violin charts, we can draw the following conclusions: Stacking ensemble learning algorithm performs best in the average accuracy and stability; GBT and DT are the second, but they still have good performance and stability. However, the performance of DT and LR is relatively poor, especially in terms of stability and overall performance.

From the analysis of Fig. 9, it can be seen that the Stacking ensemble model is significantly better than the basic learners with AUC of 0.896, DT of 0.887, SVM of 0.880 and neural network of 0.886, which shows the effectiveness of feature importance weighting and dynamic weight adjustment mechanism. It can achieve 90%+ TPR at low FPR(<0.2), especially suitable for the scene where the false alarm rate needs to be controlled in financial fraud detection; Compared with DT with the best performance, the TPR of the integrated model reaches 92.5% when FPR = 0.1, that is, it can identify 5–12% points more fraudulent transactions while maintaining the false alarm rate of 10%, showing stronger discrimination ability and practical application value.

Table 5 shows that the average single prediction delay of Stacking integrated model is significantly higher than that of single model due to the integration of multiple basic learners and the weighted calculation, especially the complex calculation brought by neural network increases the delay; At the same time, its training time is far longer than the traditional algorithm (up to 89%), because it needs to train multiple basic learners and meta-learners in stages, and involves grid search and cross-validation; In addition, the memory occupation of the model increases exponentially, and it is necessary to store multiple model parameters and intermediate results. If the deep learning model is used as the base learner, additional GPU memory support is needed. On the whole, the Stacking model relies on multi-threaded CPU parallel training, and if it includes neural network, it needs GPU to speed up the operation.

Under a unified experimental protocol (70%/15%/15% data partitioning, Accuracy/Recall/F1 evaluation, 5 repeated experiments), cross domain performance comparisons were conducted on multiple SOTA methods based on 4.2 million transaction data (including 1 million private data and 3.2 million public data including newly added IEEE-CIS Fraud, PaySim, and Alipay SEC). Table 6 shows that the Stacking method proposed in this study (integrating DT, SVM, NN, LR, combined with feature importance weighting and dynamic weighting) performs the best with 0.950 Accuracy, 0.930 Recall, and 0.940 F1; Other methods such as GNN Fraud, TabTransformer FT, LightGBM Focal, etc. have their own advantages, but their performance decreases on specific datasets due to

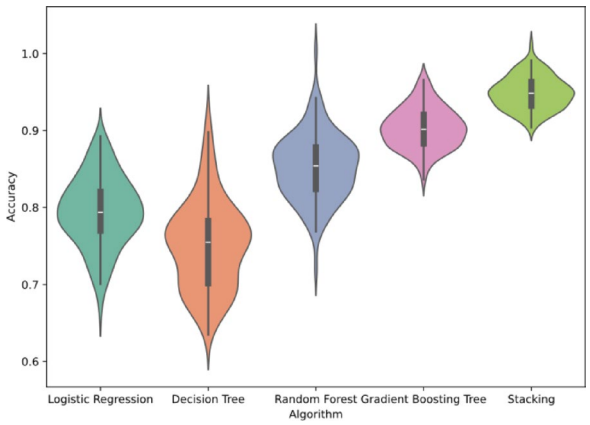


Fig. 8. Stability of performance of different models.

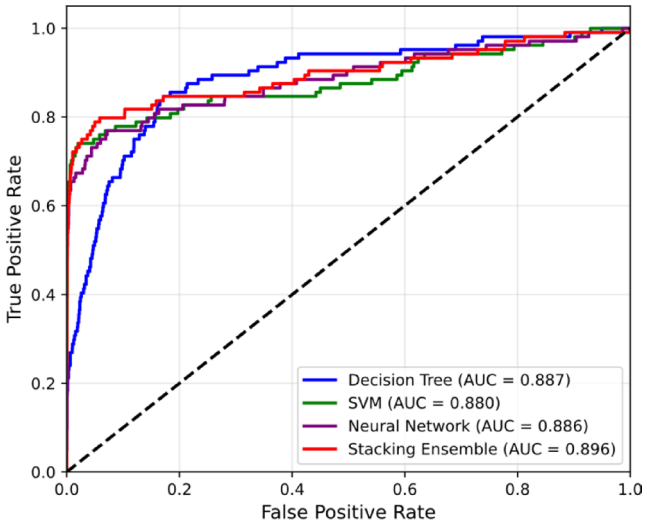


Fig. 9. AUC-ROC curve.

Model	Average single prediction delay (ms)	Training time (hours)	Memory usage (GB)	Computing resource requirements
LR	0.1	0.2	0.5	CPU (single thread)
DT	0.3	0.5	1.0	CPU (single thread)
RF	1.2	2.0	2.5	CPU (multithreading)
GBT	2.5	4.0	4.0	CPU (multithreading) + GPU (optional)
Stacking integration model	5.8 (weighted average)	8.0	7.5	CPU (multithreading) + GPU (required)

Table 5. Comparison of delay and computational cost between stacking model and other algorithms.

Method	Core model	Cross-domain average Accuracy	Cross-domain average Recall	Cross-domain average F1
Stacking (this study)	Stacking (DT + SVM + NN + LR) + feature importance weighting + dynamic weighting	0.950	0.930	0.940
GNN-Fraud	Heterogeneous GNN + Meta-learning	0.931	0.911	0.921
TabTransformer-FT	TabTransformer + Focal Loss	0.928	0.905	0.916
LightGBM-Focal	Lightgbm + focallloss + category weight	0.936	0.917	0.926
CatBoost-SMOTE	CatBoost + SMOTE + Bayesian parameter tuning	0.933	0.913	0.923
DeepGBM-FM	DeepFM + GBDT two-tower structure	0.940	0.920	0.930

Table 6. Performance comparison under unified experimental protocol.

sparse graph structures, feature drift, or high cardinality categories. This validates the robustness and leadership of the proposed method in cross domain fraud detection.

Discussion

Financial fraud is a global problem, which not only brings economic losses to the victims, but also poses a threat to the stability of the entire financial system. The fraud detection system based on Stacking integrated learning algorithm proposed in this study has high accuracy and stability, and can provide timely and accurate fraud warning for financial institutions in practical application. This helps financial institutions to find and deal with fraud in the first time, thus protecting customers' assets, maintaining market confidence and improving the overall security and customer satisfaction of financial services.

The weighting steps of feature importance and the mechanism of dynamically adjusting the weight of learners proposed in this study also provide new ideas for financial institutions to deal with complex data. These technologies can not only improve the accuracy of fraud detection, but also be applied to credit risk assessment, market trend prediction and other fields, thus comprehensively improving the risk management ability and business efficiency of financial institutions.

ML algorithm shows great potential in financial fraud detection. They can automatically extract useful features from a large number of data and find patterns and associations hidden in the data, thus realizing accurate identification of fraud. However, these algorithms also have some limitations. The performance of ML model is highly dependent on the quality and quantity of data. If the training data is biased or insufficient, the model may not be able to accurately identify all types of fraud. ML model usually takes some time to adapt to the new fraud mode. In the realistic environment where fraudulent means are constantly changing, this may cause the model to be unable to effectively deal with new fraudulent behaviors in a short time.

In practical application, although the Stacking model is excellent in accuracy, recall and F1 score, financial institutions still need to consider other important factors when adopting the model, among which model delay and calculation cost are two crucial challenges. These factors are directly related to the feasibility and efficiency of the model in the actual business environment and have a far-reaching impact on the decision-making of financial institutions. In the detection of financial fraud, the response speed of the model is very important. The model with too high delay may lead to fraud not being identified in time, thus increasing the economic losses of financial institutions and customers. Stacking model, as an integrated learning algorithm, improves the accuracy, but it may also introduce extra computing time because it needs to run multiple basic learners at the same time and integrate them in the second stage. This may lead to the delay of the model in dealing with real-time transactions, and affect the immediacy of fraud detection. When financial institutions adopt the Stacking model, they must balance the accuracy and real-time requirements of the model to ensure that the model can meet the requirements of rapid response while ensuring the accuracy.

Calculation cost is another factor that cannot be ignored. Stacking model needs a lot of computing resources in the process of training and optimization, including high-performance CPU, GPU and a lot of memory and storage space. This may lead financial institutions to invest huge costs in hardware upgrade and maintenance. In addition, with the continuous increase of data volume and the improvement of model complexity, the calculation cost may continue to rise. When financial institutions decide to adopt the Stacking model, they need to carefully evaluate their own technical infrastructure and budget constraints to ensure that they can bear the relevant calculation costs and consider the long-term operation and maintenance costs of the model.

In order to overcome the limitations of the existing fraud detection model and improve its performance, the future research can be developed in the following directions: First, the training data set is expanded and enriched by adopting more sophisticated data enhancement technology, and at the same time, the data preprocessing step is strengthened to eliminate noise and outliers, thus improving the data quality and enhancing the generalization ability of the model. Secondly, develop a dynamic model updating mechanism that can quickly adapt to new fraudulent means, and use online learning, incremental learning or migration learning to ensure that the model maintains high efficiency in the ever-changing fraudulent environment. Further, we can explore the integration of data from different modes (such as text, image and sound) to obtain a more comprehensive perspective of fraud detection and capture the characteristics of fraud more accurately. Finally, strengthening the interpretive approach of ML model will make financial institutions understand the decision logic of the model more deeply, which will not only enhance the trust of institutions in the model, but also promote its application and promotion in business. In the future, we will explore the use of Synthetic Minority Over-sampling Technique (SMOTE) and Generative Adversarial Networks (GAN) to increase the number of rare fraud cases, thereby enhancing the model's capability to handle imbalanced datasets. We will also investigate online learning and incremental learning methods to enable the model to adapt in real-time to new fraud patterns and market changes, maintaining its predictive accuracy.

Conclusion

Through in-depth analysis and comparison of the application effects of different ML algorithms in financial fraud detection, this study found a method that can significantly improve the accuracy of fraud detection. In particular, Stacking ensemble learning algorithm is the best in this study, which is significantly better than other algorithms in accuracy, recall and F1 score. This is due to the fact that Stacking algorithm can make full use of the advantages of multiple basic learners, and further improve the performance of the model through the weighting step of feature importance and the mechanism of dynamically adjusting the weights of learners. However, the ML method has the problem of performance degradation when dealing with high-dimensional data and unbalanced data sets, and some complex models, such as deep learning network, have strong performance but poor explanatory power, which requires high data and computing resources. In addition, the continuous updating and changes of financial fraud means require that the detection model can adapt to the new situation quickly, and the existing methods still need to be improved in this respect.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author Jing Jin on reasonable request via e-mail jinj@sumhs.edu.cn.

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Author contributions

Jing Jin: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation Yongqing Zhang: writing—review and editing, visualization, supervision, project administration, funding acquisition.

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Competing interests

The authors declare no competing interests.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. All methods were performed in accordance with relevant guidelines and regulations.

Additional information

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